

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS

MONDAY, 15 MAY 2023

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONS)
BACHELOR OF INFORMATION SYSTEMS (HONS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY

Instructions to Candidates:

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **CHOOSE ANY ONE (1)** question in **Section B**.
Each question carries 25 marks each.

Note: Should you answer more than ONE (1) question in section B, marks will only be awarded to the FIRST question attempted.

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS**Section A (Answer ALL Questions)**

- Q1. (a) Table 1.0 shows the sequences of activities in UTAR's project for software development.

***Note: The value of S = 5.**

Activity	Expected Time	Predecessor
1. Requirement Collection	4S-S	-
2. Requirement Analysis	3S	1
3. Prototype	3S+S	2
4. Programming & Testing	50S/2S	2,3
5. Installation	3S – (S-2)	4

Table 1.0: Sequence of activities in UTAR's project.

- (i) Calculate the Early Start (ES) values for the activities listed in Table 1.0. (5 marks)
- (ii) Calculate the Early Finish (EF) values for the activities listed in Table 1.0. (5 marks)
- (b) **Project charter** is a one-page document that provides high level overview of a software project development to the key stakeholders during the project initiation phase. Discuss **THREE (3)** advantages of a project charter. (6 marks)
- (c) In an economic feasibility study of XYZ software development project the overall Net Present Value (NPV) cash flow is observed as RM 200,056, for five years. While the break-even occurs in 2.20 years. In your opinion, state if the XYZ software project is generating profit or otherwise. Justify your answer. (6 marks)
- (d) **Reuse** is the use of previously written software resources in new software development. Provide **THREE (3)** reuse approaches. (3 marks)
- [Total : 25 marks]

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- Q2. (a) Support service is important for system users. The primary goal of system support services is to ensure the system runs smoothly, efficiently, minimizing downtime, and maximizing productivity.
- (i) Explain **TWO (2)** types of automated system support services. (6 marks)
- (ii) **Help desk** is a single point for all user enquiries about a particular system and is staffed by Information System (IS) personnel. These IS personnel are trained before they could assist in user enquiries. In your opinion, discuss why training is important for the IS personnel in help desk. (8 marks)
- (b) In a system implementation, the efforts from the development team do not necessarily to ensure the success of the system. Deduce **FOUR (4)** major factors that influence the system implementation success. (4 marks)
- (c) Indicate **ONE (1)** information system that you had designed in your assignment. As a developer of the system, explain how performance testing shall determine if the system meets the users' requirements. (7 marks)
- [Total : 25 marks]

- Q3. (a) System maintenance refers to the process of improving a system to run efficiently without bugs or errors. There are four types of system maintenances.
- (i) Discuss the differences between corrective and preventive maintenances. (8 marks)
- (ii) A private company has a payroll system that has been in use for several years. The system was developed with a focus on functionality. However, over time, the employees whom are the users of the payroll system have identified and suggested potential areas to improve the system. Therefore, the company has decided to perform maintenance to the system. As a system maintenance personnel for the company, you have studied and analyzed the potential improvement areas suggested by the employees. With that, you have decided to perform adaptive and perfective maintenances to the payroll system. Justify to the company why you must perform adaptive and perfective maintenances to the mentioned system. (8 marks)
- (b) In a study, it was found that the cost of system maintenance is dependent on its maintainability.
- (i) Suggest **THREE (3)** factors that influence the decreasing cost of system maintenance in an organization. (3 marks)

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS**Q3. (b) (Continued)**

- (ii) A software development company has received multiple requests for system maintenances from their clients. Discuss how the company should control and attain to these maintenance requests by providing timely support to the clients. (6 marks)

[Total : 25 marks]

Section B (Choose any ONE question)

- Q4. (a) Figure 1.0 shows a basic sketch of a customer's ArnaX online banking dashboard user interface (UI) using a computer aided software engineering (CASE) tool.

+-----+		
ArnaX Online Banking		
Dashboard		
+-----+		
Account Details		
+-----+		
Account Number: XXXX-XXXX-XXXX-1234		
Account Type: Savings		
Available Balance: RM 1,200.00		
+-----+		
Transaction History		
+-----+		
Date	Description	Amount
+-----+		
02/12/2023	Grocery Store	-RM 50.00
02/01/2023	Employer	RM 1,200.00
01/31/2023	Rent	-RM 800.00
01/30/2023	Internet	-RM 50.00
01/28/2023	Refund	RM 200.00
+-----+		
Account Services		
+-----+		
View Statement Update Profile		
+-----+		

Figure 1.0: Sketch of a customer's ArnaX online banking dashboard user interface (UI).

Referring to Figure 1.0, prepare the items listed below to perform normalization using third normal form (3NF).

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS**Q4. (a) (Continued)**

- (i) Determine **THREE (3)** relations from Figure 1.0. (4 marks)
 - (ii) Determine **First Normal Form (1NF)** for the relations identified in Q4(a)(i). (3 marks)
 - (iii) Determine **Second Normal Form (2NF)** for the 1NF relations identified in Q4(a)(ii). (3 marks)
 - (iv) Determine **Third Normal Form (3NF)** for the 2NF relations identified in Q4(a)(iii). (3 marks)
 - (v) Determine if **associative entity(s)** exist for the 3NF relations identified in Q4(a)(iv). Justify your answer. (6 marks)
- (b) System requirements determination is divided into traditional and contemporary methods. State **THREE (3)** traditional methods for determining system requirements. (6 marks)
- [Total : 25 marks]

- Q5. (a) In a retail clothing website, the customers can browse for specific clothes using stock keeping unit (SKU). Moreover, the customers are able to search for clothes by category such as cloth type, size, color, or price range. If the customers have selected the desired clothing, the customer is directed to the checkout page. Otherwise, the customers can continue browsing or remove the clothing from their cart. At the checkout page, the customers with membership card shall enter their membership number to earn rewards or discounts. While the customers without a membership card is required to become a member to purchase the selected clothing. The customers with membership are directed to make the full payment including the shipping fees upon checkout. In addition, the customers shall receive a receipt for their purchase(s) via email and updates on the order delivery status for successful payment. However, the transaction shall end displaying an error page for any unsuccessful payment.

According to the above scenario, prepare the items listed below to form a decision tree.

- (i) Determine **THREE (3)** conditions for the given scenario. (3 marks)
- (ii) Determine **FOUR (4)** actions for the given scenario. (4 marks)
- (iii) Draw a decision tree of **FOUR (4)** rules with the listed conditions in Q5(a)(i) and the actions identified in Q5(a)(ii). (8 marks)

UCCD1013 ANALYSIS AND DESIGN OF INFORMATION SYSTEMS**Q5. (Continued)**

- (b) **User interface (UI)** is a method for users to interact with a system. Deduce **THREE (3)** approaches for UI interaction. (6 marks)
- (c) **Computer mouse** is a common hardware device used for user interface (UI) interaction. Explain **ONE (1)** usability limitation while using a computer mouse for UI interaction. (4 marks)

[Total : 25 marks]